

Technical Report

1. Research Question

This project asks: Does participation in a job-training program cause an increase in later earnings? The treatment variable is job-training participation, and the outcome variable is post-program earnings.

2. Data

The cleaned analysis sample contains 16,177 observations. There are 185 treated observations and 15,992 control observations. The main control variables are age, education, race indicators, marital status, degree status, and prior earnings.

3. Prediction vs. Causation

A prediction-only approach is insufficient because it would estimate who is likely to earn more, not whether training itself caused the earnings difference. The policy question is counterfactual: what would happen to earnings if an individual received training instead of not receiving training?

4. Identification Strategy

The main identification strategy is Double Machine Learning. DML first estimates the relationship between controls and the outcome, then estimates the relationship between controls and treatment assignment. The final treatment effect is estimated using residualized variation after accounting for observed confounders.

5. Results

The naive OLS estimate is \$-8,498. The adjusted OLS estimate is \$699. The Double Machine Learning estimate is \$839, with a 95 percent confidence interval of [\$-556, \$2,235]. The DML standard error is \$712, and the p-value is 0.2383.

6. Interpretation

The DML estimate suggests that the estimated effect of job training on later earnings is \$839 for the analysis sample. However, this estimate should be interpreted as causal only if the conditional independence assumption is credible. If important unobserved factors affect both training participation and earnings, the estimate may still be biased.

7. Robustness

Comparing naive OLS, adjusted OLS, and DML helps show the role of observed covariates. The naive estimate of \$-8,498 does not control for baseline differences. The adjusted estimate of \$699 and the DML estimate of \$839 provide more credible estimates under the selection-on-observables framework.

8. Conclusion

The project provides a causal estimate of the relationship between job training and post-program earnings. The main result is an estimated effect of \$839, with a 95 percent confidence interval of [\$-556, \$2,235]. The recommendation is to treat the program as potentially valuable but to continue improving data collection and evaluation design.